

NMDC Steel ET — Non-Technical Awareness Revision Sheet

Company · Mining · Drilling & Blasting · Steel Making · Rubber & Materials · Industrial Safety
For the general aptitude / awareness section of the Executive Trainee CBT (1 October 2026). Recall-level coverage only.

1. NMDC & NMDC Steel Limited — Company Awareness

Point	Detail
NMDC full form	National Mineral Development Corporation Limited
Incorporated	1958
Ministry	Ministry of Steel, Government of India
Status	Navratna Public Sector Enterprise (PSE)
Headquarters	Hyderabad, Telangana
Core business	India's largest iron ore producer & exporter; also diamond mining
Key iron ore mines	Bailadila sector (Chhattisgarh), Donimalai & Kumaraswamy (Karnataka)
Diamond mine	Panna, Madhya Pradesh — India's only mechanised diamond mine
NMDC Steel Ltd (NSL)	Demerged from NMDC as a separate listed company; owns the steel plant
Nagarnar plant	3.0 MTPA Integrated Steel Plant, Nagarnar, Bastar district, Chhattisgarh
This recruitment	Notification No. 05/2026 — 102 Executive Trainee posts, 10 disciplines
Posting	Nagarnar Integrated Steel Plant, Chhattisgarh
Selection	CBT 70% + Group Discussion 15% + Interview 15%; GD/PI shortlist ratio 1:3

Likely question types: which state hosts Bailadila / Nagarnar / Panna; NMDC's Navratna status; which ministry NMDC falls under; what NMDC primarily mines; headquarters city.

Related organisations worth recognising

- **SAIL** — Steel Authority of India Ltd, Maharatna; plants at Bhilai, Bokaro, Rourkela, Durgapur, Burnpur (IISCO), Salem.
- **DGMS** — Directorate General of Mines Safety; the mine safety regulator, under Ministry of Labour & Employment. HQ Dhanbad.
- **IBM** — Indian Bureau of Mines (Nagpur), mineral conservation & regulation.
- **GSI** — Geological Survey of India (Kolkata), mineral exploration.
- **PESO** — Petroleum & Explosives Safety Organisation (Nagpur), licences explosives.
- **MECON, RINL/Vizag Steel, KIOCL, MOIL, Coal India** — other steel/mining PSUs.

2. Iron Ore & Mining Fundamentals

Iron ore minerals (high-frequency question)

Ore	Formula	Fe content	Note
Haematite	Fe ₂ O ₃	~70% (max)	Reddish; the main Indian ore; Bailadila is high-grade haematite
Magnetite	Fe ₃ O ₄	~72% (highest)	Black, magnetic; largest Indian reserves in Karnataka
Limonite	hydrated oxide	~60%	Yellow-brown, inferior
Siderite	FeCO ₃	~48%	Carbonate, lowest grade

Trap: magnetite has the *highest iron percentage*, but haematite is the *most important/most used* ore in India. Questions exploit this difference.

Mining methods

- **Open-cast (surface) mining** — shallow deposits; cheaper, safer, higher recovery, large-scale mechanisation. Most Indian iron ore including Bailadila. Involves removing **overburden** (waste rock above the ore).
- **Underground mining** — deep deposits; costlier, higher risk (gas, roof fall, ventilation), lower recovery. Methods: room-and-pillar, longwall, block caving, stoping.
- **Placer / alluvial mining** — from river/stream deposits.
- **Stripping ratio** = overburden removed ÷ ore extracted. Lower is better/more economical.
- **Bench** — the horizontal step/level in an open-cast mine. **Berm** — safety ledge. **Haul road** — dumper route.

Unit operations sequence

Exploration → Drilling → Blasting → Loading (shovel/excavator) → Haulage (dumper/conveyor) → Crushing → Screening → Beneficiation → Pelletisation/Sintering → Dispatch

Mineral processing terms

- **Beneficiation** — upgrading ore by removing gangue (waste minerals) to raise Fe%.
- **Comminution** — size reduction: crushing (coarse) then grinding (fine).
- **Screening** — size separation. **Classification** — separation by settling in fluid.
- **Froth flotation** — separation using surface-property differences and air bubbles.
- **Magnetic separation** — used for magnetite. **Gravity separation** (jigs, spirals) — density-based.
- **Pelletisation** — fine concentrate rolled into spherical pellets (~9–16 mm) and hardened; feed for blast furnace/DR.
- **Sintering** — fines fused into porous lumps with flux and coke breeze.
- **Tailings** — waste slurry after processing; stored in tailing dams.
- **Slime** — very fine waste fraction. **ROM** — Run of Mine (as-mined ore).

Mining law & regulation

- **Mines Act, 1952** — principal law on health, safety and welfare of mine workers; administered via DGMS.
- **MMDR Act, 1957** — Mines and Minerals (Development and Regulation) Act; leases, royalties, auctions.
- **Metalliferous Mines Regulations, 1961** — operational rules for non-coal mines (relevant to iron ore).
- **Explosives Act, 1884 & Explosives Rules, 2008** — govern manufacture, storage, transport and use of explosives; licensing by PESO.
- **Statutory posts:** Manager, Agent, Owner, Mining Engineer, Surveyor, Shotfirer/Blaster, Safety Officer — each requires a DGMS-recognised certificate of competency.

3. Drilling & Blasting — Terminology and Safety

This section is **definition and safety-regulation level only**, which is all the exam tests. Questions in PSU papers ask what a term means, who is legally authorised, and which safety rule applies — never how to prepare or charge an explosive.

Why blasting is used

Hard rock and high-grade ore cannot be economically excavated by machine alone. Controlled blasting fragments the rock mass so that shovels and crushers can handle it. The objective is **correct fragmentation size** with minimum ground vibration, flyrock, noise and back-break.

Core terminology (memorise these)

Term	Meaning
Blast hole	Drilled hole in which the charge is placed
Burden	Distance from the blast hole to the nearest free/open rock face
Spacing	Distance between adjacent blast holes in the same row
Bench height	Vertical height of the working level being blasted
Sub-drilling (sub-grade)	Drilling slightly below the intended floor level to avoid a raised toe
Stemming	Inert material (drill cuttings, sand) filling the top of the hole to confine energy downward
Collar	Top/mouth portion of the blast hole
Toe	Bottom of the bench face; "toe problem" = unbroken rock left at floor level
Primary blasting	The main blast that breaks rock from the solid face
Secondary blasting	Breaking oversized boulders produced by the primary blast
Delay blasting	Firing holes in timed sequence (milliseconds apart) to improve fragmentation and cut vibration
Misfire	A charge that fails to detonate — a major hazard; only the authorised shotfirer may deal with it, under regulation
Flyrock	Rock thrown beyond the intended zone — a leading cause of blasting accidents
Air blast / overpressure	Pressure wave in air causing noise and window damage
Ground vibration	Seismic effect on nearby structures; measured by seismograph in mm/s (PPV)
Powder factor	Explosive consumed per unit of rock broken (kg per tonne or per m ³) — an efficiency index
Blast pattern	Geometric layout of holes — e.g. square, rectangular, staggered
Controlled blasting	Techniques used near sensitive structures to limit damage: pre-splitting, smooth blasting, line drilling, cushion blasting

Classes of commercial blasting agents (names only)

- **Permitted explosives** — specially approved for gassy coal mines.
- **Slurry / water-gel and emulsion explosives** — water-resistant, widely used in wet holes; considered safer to handle than older nitroglycerine-based types.
- **ANFO** — the common bulk agent for dry holes; not water-resistant.
- **Initiation devices** — detonators (plain, electric, non-electric/shock-tube, electronic), detonating cord, safety fuse, and boosters. Electronic detonators give the most precise delays.
- **Storage** — explosives and detonators must be kept in separate licensed magazines, with strict accounting of issue and return.

Blasting safety procedure — the exam-relevant sequence

- Blasting only by a **competent, authorised shotfirer/blaster** holding a valid certificate.
- The area is cleared of all persons and machinery; approaches are guarded by posted sentries.
- **Audible warning signals** are sounded before blasting, and a distinct **all-clear signal** after.
- Personnel take cover in designated **blast shelters** at a prescribed safe distance.
- Re-entry only after the statutory waiting period, once fumes and dust have cleared and the face has been inspected by the shotfirer.
- Misfires are reported, recorded, and handled strictly per regulation — never by drilling into a charged hole.
- Day-blasting preference, no blasting during electrical storms, and a full account of explosives used vs returned.

Most probable MCQs: definition of burden / spacing / stemming / powder factor; purpose of delay blasting; what a misfire is and who may handle it; which agent is not water-resistant; which Act governs explosives; who issues explosives licences (PESO); what flyrock is; purpose of secondary blasting.

4. Iron & Steel Making

The two production routes

Feature	BF-BOF route	DRI-EAF route
Sequence	Blast Furnace → Basic Oxygen Furnace	Direct Reduced Iron (sponge iron) → Electric Arc Furnace
Reductant	Coke (from coking coal)	Natural gas or non-coking coal
Product of 1st stage	Hot metal / pig iron (liquid, high carbon)	Sponge iron (solid, porous)
Scale	Large integrated plants	Smaller/secondary producers
Note	Nagarnar is an integrated steel plant	India is the world's largest sponge iron producer

Blast furnace essentials

- **Charge (burden)**: iron ore (lump/sinter/pellets) + coke + flux (limestone/dolomite).
- **Coke** does three jobs: fuel, reducing agent, and permeable support for the burden.
- **Flux** (limestone, CaCO_3) removes impurities as slag.
- **Hot blast** of preheated air is injected through **tuyeres**; preheated in **stoves/regenerators**.
- Main reducing gas is **CO**; the overall process is **reduction** of iron oxide.
- Outputs: **hot metal** (tapped from the hearth), **slag** (floats above, used in cement), **BF gas** (low calorific value fuel).
- Zones top to bottom: stack/shaft → belly → bosh → hearth. **Tap hole** for metal, **slag notch** for slag.
- **Pig iron** contains roughly 4% carbon; steel has under ~2% (practically under 1.5%); wrought iron is the purest form of iron.

Steel making & downstream

- **BOF / LD converter** — pure oxygen is blown into hot metal to oxidise excess carbon; "basic" refers to the basic refractory lining. Very fast heat times.
- **EAF** — melts scrap/sponge iron using electric arcs from graphite electrodes.
- **Secondary refining / ladle metallurgy** — LF, VD/VOD, RH degassing for precise composition.
- **Continuous casting** — liquid steel cast directly into **slabs, blooms, billets**; replaced ingot casting; higher yield.
- **Hot rolling** — above recrystallisation temperature; no work hardening, scaled surface, cheaper.
Cold rolling — below it; better surface finish, tighter tolerance, increases strength/hardness by work hardening.
- Products: HR coil, CR coil, plates, wire rod, TMT bars, galvanised sheet.
- **By-products**: slag (cement, road base), BF/coke oven gas (fuel), tar, ammonium sulphate, benzol.

Policy & green steel

- **National Steel Policy 2017** — targets 300 MT crude steel capacity by 2030-31.
- India is the **second largest crude steel producer** in the world (China first).
- **Green steel** — hydrogen-based direct reduction replacing coke; also scrap-based EAF, CCUS, renewable power. Steel is among the hardest sectors to decarbonise.
- Steel plants are energy- and water-intensive; key environmental concerns are particulate emissions, CO_2 , and slag/tailings disposal.

5. Rubber — Material Basics and Plant Uses

Your mock asked about "use of rubber". In a mine/steel-plant context this is tested two ways: rubber as an **engineering material**, and rubber as an **electrical-safety insulator**. Cover both.

Material basics

- **Natural rubber** — obtained as **latex** from the *Hevea brasiliensis* tree; polymer of **isoprene**. In India, mainly Kerala.
- **Vulcanisation** — heating rubber with **sulphur** to cross-link the chains. Discovered by **Charles Goodyear**. Result: greater strength, elasticity, hardness, abrasion resistance, and much lower sensitivity to temperature. Very commonly asked.
- **Synthetic rubbers** — e.g. neoprene (oil/chemical resistant), SBR (tyres), nitrile (oil resistant), butyl (air-tight), silicone (heat resistant), EPDM (weather resistant).
- Key properties: **elasticity, abrasion resistance, vibration and shock absorption, water resistance, and excellent electrical insulation**.
- Rubber is an **elastomer**; it is an **insulator** (very low electrical conductivity) — this is the basis of its safety use.

Uses in mining and steel plants

- **Conveyor belts** — the single biggest use; rubber belts move ore, coke, sinter and pellets. Rubber gives flexibility plus abrasion and impact resistance.
- **Rubber lining** of chutes, hoppers, pipes, pumps and cyclones to resist abrasive slurry wear and reduce noise.
- **Rubber mill liners** in grinding mills — lighter and quieter than steel liners.
- **Screen panels and decks** in screening plants.
- **Tyres** on dumpers, loaders, graders — rubber-tyred haulage.
- **Anti-vibration mounts and expansion joints** to isolate machinery vibration.
- **Seals, gaskets, O-rings, diaphragms, hoses** throughout plant piping.
- **Flotation and slurry hose** handling corrosive/abrasive media.

Rubber as safety equipment (high-probability MCQ area)

- **Electrical insulating gloves** — rubber gloves for work on or near live equipment; tested to a voltage class.
- **Rubber mats / insulating matting** in front of switchboards, panels and substations — standard requirement in electrical rooms.
- **Rubber safety boots / gum boots** — electrical insulation plus protection from water, slurry and chemicals.
- **Rubber aprons and sleeves** for acid/chemical handling (pickling lines).
- **Insulating rods, blankets and line hose** used in electrical maintenance.
- Rule of thumb for MCQs: if the question is "which material is used for protection against **electric shock**", the answer is rubber (or another insulator such as PVC/ebonite), never a metal.

6. Industrial Safety — Highest-Yield Section

Core definitions

- **Hazard** — a source or condition with the *potential* to cause harm. **Risk** — the *likelihood* × *severity* of that harm occurring. Distinguishing these is a classic MCQ.
- **Accident** — an unplanned event causing injury/damage. **Incident** — may or may not cause harm. **Near miss** — no injury occurred but could have; must still be reported.
- **Unsafe act** (human behaviour) vs **unsafe condition** (physical state). Most accidents trace to unsafe acts.
- **Heinrich's accident triangle** — many near misses and minor injuries underlie each major injury; also the domino theory of accident causation.
- **Frequency rate** and **severity rate** — standard injury statistics per million man-hours.

Hierarchy of hazard control — memorise the order

Elimination → Substitution → Engineering controls → Administrative controls → PPE

PPE is always the **last** line of defence, never the first. Questions frequently test this.

PPE — match the hazard

PPE	Protects against
Safety helmet / hard hat	Falling objects, head impact
Safety goggles / face shield	Flying particles, chemical splash, arc and molten metal splash
Ear plugs / ear muffs	Noise (muffs give higher attenuation than plugs)
Respirator / dust mask	Dust, fumes, gases; SCBA for confined space or toxic atmosphere
Safety shoes (steel toe)	Falling/rolling objects, puncture
Rubber gloves / mats / boots	Electric shock (insulation)
Leather gloves / apron	Heat, sparks, welding, hot metal handling
Full body harness / lifeline	Fall from height (mandatory above ~1.8 m / as per rule)
Reflective jacket	Visibility near moving heavy equipment
Welding shield with filter lens	Arc radiation, UV/IR

Safety sign colour coding (very commonly asked)

Colour	Meaning	Shape / example
Red	Prohibition, danger, stop; fire-fighting equipment	Circle with diagonal bar — "No smoking"
Yellow / amber	Caution, warning, risk of danger	Triangle — "Danger: high voltage"
Blue	Mandatory action / must be obeyed	Solid circle — "Wear your helmet"
Green	Safe condition, first aid, emergency exit	Square/rectangle — "Exit", "First aid"

Fire — classes and correct extinguishers

Class	Fuel	Suitable extinguishing agent
A	Ordinary solids — wood, paper, cloth	Water, foam
B	Flammable liquids — petrol, oil, paint	Foam, CO ₂ , dry chemical powder (DCP) — not water
C	Flammable gases (and in some systems, electrical)	DCP, CO ₂ ; isolate the gas supply
D	Combustible metals — magnesium, sodium	Special dry powder only — never water
E / electrical	Live electrical equipment	CO ₂ , DCP — never water or foam
K / F	Cooking oils and fats	Wet chemical

- **Fire triangle** = fuel + oxygen + heat. Adding the chain reaction gives the **fire tetrahedron**. Extinguishing = removing one element (cooling, smothering, starving, inhibiting).
- **PASS** = Pull, Aim, Squeeze, Sweep — extinguisher operation. Aim at the **base** of the fire.
- **Flash point** — lowest temperature at which a liquid gives off vapour that ignites momentarily. Lower flash point = more hazardous.

Permit and isolation systems

- **Work permit system** — written authorisation before hazardous jobs. Types: hot work, cold work, confined space entry, electrical, height, excavation.
- **LOTO (Lock Out - Tag Out)** — energy sources isolated, locked, and tagged before maintenance. Purpose: prevent accidental energisation. Only the person who applied the lock removes it.
- **Confined space entry** — requires atmospheric testing (oxygen, flammables, toxics), continuous ventilation, a standby attendant outside, harness and lifeline, and a rescue plan. Hazards: oxygen deficiency, toxic gases, engulfment.
- **Hot work** — welding/cutting near flammables; requires gas testing, fire watch, and screens.

Assessment and management tools

- **HIRA** — Hazard Identification and Risk Assessment. **JSA** — Job Safety Analysis. **HAZOP** — Hazard and Operability study. **FMEA**, **Fault Tree Analysis**.
- **Safety audit** — systematic review of safety systems. **Toolbox talk** — short pre-job safety briefing.
- **MSDS / SDS** — Material Safety Data Sheet; gives properties, hazards, handling, storage, first aid and emergency measures for a chemical.
- **Mock drill** — practice of the emergency/on-site emergency plan.
- **ISO 45001** — occupational health & safety management standard (replaced OHSAS 18001). **ISO 9001** quality, **ISO 14001** environment.
- **5S** — Sort, Set in order, Shine, Standardise, Sustain. Housekeeping reduces accidents.

Legislation

- **Factories Act, 1948** — health, safety, welfare, working hours in factories; provisions on guarding of machinery, ventilation, lighting, safety officers.
- **Mines Act, 1952** — the mining counterpart.
- **Workmen's/Employee's Compensation Act, ESI Act, Environment Protection Act 1986, Air Act 1981, Water Act 1974.**
- **National Safety Day** — 4 March (National Safety Council of India founded that date). **World Environment Day** — 5 June.

Plant-specific hazards worth recognising

- **Steel plant:** molten metal splash, hot surfaces, CO poisoning from BF/coke oven gas, noise, heat stress, dust/silica, crane and material-handling accidents, gas leaks.
- **Mine:** flyrock and misfires, slope/bench failure, dumper and haul-road accidents, dust (silicosis, pneumoconiosis), noise-induced hearing loss, inundation, tailing dam failure.
- **Occupational diseases:** silicosis (silica dust), pneumoconiosis (coal dust), asbestosis, NIHL (noise), heat stroke, vibration white finger.

Exam-day reminder: reports indicate 60 discipline questions + 40 general aptitude with no negative marking — so attempt every single question and leave nothing blank. Confirm the marking scheme on your admit card instruction page before you start, since the official notification did not specify it.